Local-Global Context-Aware Generative Dual-Region Adversarial Networks for Remote Sensing Scene Image Super-Resolution

Huiting Li, Weihuan Deng, *Graduate Student Member, IEEE*, Qiqi Zhu, *Senior Member, IEEE*, Qingfeng Guan, and Jiancheng Luo

***Abstract*— Recently, high-resolution (HR) remote sensing images (RSIs) have attracted increasing attention in a number of tasks. Super-resolution (SR) is an efficient method to obtain HR RSIs. Due to the influence of imaging distances and angles, RSIs significantly differ from natural images in terms of land cover element distribution, ground object scale, and scene complexity. This poses a challenge for capturing global and local low and high-frequency and restoring fine image details for RSI SR. In this article, a local-global context-aware generative dual-region adversarial network (LGC-GDAN) is designed for RSI SR. It is composed of dual region-level discriminators and a dual-path generator with a context-aware network and an edge-assisted network. To capture global and local low and high-frequency information, the global-aware self-attention (GAS) mechanism and local-aware self-attention (LAS) mechanism are introduced into the context-aware network. The GAS mechanism combines high-pass and low-pass filtering for long-range similarity feature, while LAS uses local aggregation for fine-level feature. The low-resolution (LR) images and the corresponding edge maps are input to the edge-assisted network to extract the detailed geometric structure. To address small ground object and complex ground scenes, conventional image-level discriminators exhibit limited performance in capturing detailed information. Unlike previous discriminator, a region-level discriminator is designed to obtain the real/fake label of each local region. Moreover, two task-driven loss functions are designed to produce diverse images for further scene classification. The experiments undertaken on several remote sensing datasets demonstrate that LGC-GDAN outperforms the other state-of-the-art methods.**

***Index Terms*— Generative adversarial network (GAN), high-resolution (HR), scene classification, super-resolution (SR).**

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I. Introduction

With the rapid development of satellite imaging techniques over recent years, the spatial resolution of satellite imagery has been enhanced and has a wide range of applications in the military and civilian domain, such as natural disaster detection, urban functional analysis, and environment monitoring [1], [2], [3], [4]. Unfortunately, the cost of using such high-resolution (HR) imagery can make it impractical when large areas have to be covered or if the multitemporal analysis has to be undertaken [5], [6], [7]. In this field, the open-source data with acceptable spatial quality can be considered, like those provided by satellites such as Landsat or Sentinel [8].

Since the spatial details of open-source data may not be sufficient for some remote sensing applications [9], [10], [11], there is an urgent need to develop a practical and effective algorithm to overcome the shortcomings of remote sensing images (RSIs) and save a lot of money [12]. Single image super-resolution (SISR) has become a hot topic in the area of image processing and is widely used to improve the spatial resolution of images [13], [14], [15]. However, the ill-posed problem is the biggest challenge for SISR since any low-resolution (LR) input may correspond to multiple HR images [16]. For this reason, much more HR pixels need to be estimated from LR pixels, especially in the large desired scale factor case, this problem is more severe [17], [18].

Deep convolutional neural networks (CNNs) have been attracting more attention due to their strong feature learning ability in various computer vision tasks, such as image classification, semantic segmentation, and object detection [19], [20], [21], [22]. Up until now, numerous solutions are proposed to improve the performance of the super-resolution (SR) models [22], [23]. CNN-based SISR methods have achieved good performance on various publicly available datasets [24]. These methods try to establish the mapping relationship between LR and HR image pairs. Nevertheless, CNN-based methods are limited as they are defined based on pixel-wise difference, which results in the recovered images being perceptually unsatisfactory [25]. To obtain pleasing SR results in terms of both visual appearance and local details, generative adversarial network (GAN)-based methods [26], [27] are proposed to produce HR images with higher perceptual quality. However,

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most of these learning-based SR methods are elaborately designed for natural images [28]. Compared with natural images, RSIs have a complicated background, diverse ground objects, and weak texture [29]. These factors will lead to inconsistent mapping relationships between HR-RSIs and LR-RSIs [30].

To improve the reconstruction performance of the SR model on RSIs, a local-global combined network (LGCNet) [31] is designed to learn the local details and global priors by leveraging the multilevel data representation ability of the deep neural networks. To compensate for the missing details of the SR process, a mixed high-order attention network (MHAN) [32] with feature extraction and feature refinement components is proposed to leverage hierarchical features at different levels. Since CNN-based SR methods are difficult to obtain perceptually pleasing results, the improved PatchGAN [33] is proposed and the novel loss function is designed to restore abundant detailed information. Considering that RSIs have more low-frequency components than natural images, a dual pathway network is specifically designed to improve the discrimination ability and address the problem of "discrimination-ambiguity" [34]. In addition, noise often appears in remote sensing satellite images and interferes with the recovery of high-frequency information. A GAN-based edge-enhancement network (EEGAN) is proposed to purify the noise-contaminated regions in the RSIs and produce esthetically pleasing results [27]. To make full use of the historical RSIs for SR reconstruction, a reference-based SR model [35] is designed to compensate for the lost details in LR-RSIs by introducing additional HR-RSIs. Currently, many scholars have focused on combining image SR with other vision tasks. Shermeyer et al. have proved that using image SR technology as the preprocessing of satellite imagery can effectively improve the performance of object detection [64]. Wang et al. proposed a novel multiclass cyclic SR GAN to benchmark image SR-based object detection [65]. Moreover, the GAN-based end-to-end multitask learning architecture is proposed to simultaneously achieve SR reconstruction and semantic segmentation tasks for RSIs [66].

Although deep-learning-based SR methods have been widely applied for RSI SR, there still remains a significant challenge. Due to the influence of imaging angle and distance, RSIs are significantly different from natural images in terms of land cover element spatial distribution, ground object scale, and scene complexity. Due to different imaging angles, compared with natural images, RSIs rarely show an obvious up-down and back-forth spatial direction relationship and mainly contain more land cover elements and complex spatial distribution. As illustrated in Fig. 1(a), the high-frequency information in the red boxes, such as texture information and the low-frequency information in the green boxes, such as color information, all present short and long-distance spatial distribution phenomena. Therefore, how to design a network to fully capture important local and global low and high-frequency information in complex spatial distribution scenarios is crucial for the task of RSI SR. Compared with natural images, RSIs display smaller ground objects and more complex scenes due to different imaging distance. According

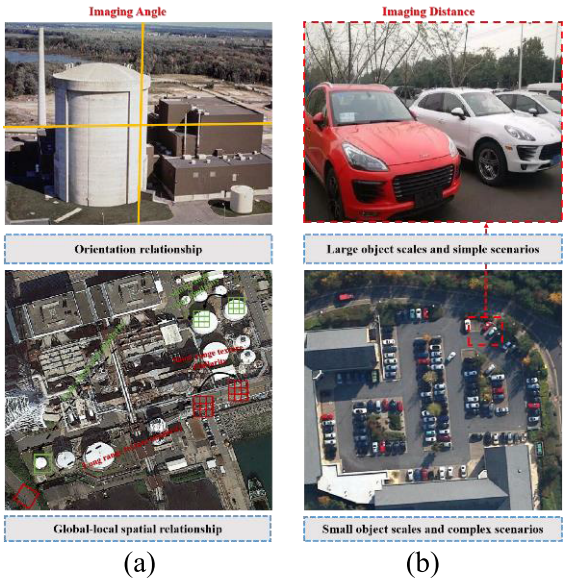

Fig. 1. Two main research motivations. (a) Spatial distribution differences caused by imaging angle. (b) Ground object scale and scene complexity difference caused by imaging distance.

to Fig. 1(b), it can be seen that the red and white cars in the natural image only occupy a small area in the RSI, while the RSI shows a more complex distribution of ground elements. Previous image-level discriminators are directly applied to RSIs and may not be able to fully capture the rich detailed information of RSIs. Hence, how to design a network to effectively achieve regional-level detailed information restoration of RSIs is also challenging.

To address the above problems, the local-global context-aware generative dual-region adversarial network (LGC-GDAN) is proposed to facilitate high-quality image SR. LGC-GDAN is constructed of dual discriminator networks and dual path generator network with a context-aware branch and edge-assisted branch, where the context-aware branch includes global-aware self-attention (GAS) mechanism, residual-in-residual dense blocks (RRDBs) [49], and local-aware self-attention (LAS) mechanism. The edge-assisted branch employs the encoder-decoder network to remove noise interference and facilitate edge information extraction. The features from the context-aware branch and edge-assisted branch are concatenated to generate high-quality SR images. With regard to discriminator, dual region-level discriminators are adopted to process the original images and corresponding edge map, respectively. Due to the inherently ill-posed problem of the SISR methods, two task-driven loss functions are designed to generate diverse SR images and used for subsequent scene classification. Overall, the main contributions of this article are summarized as follows.

1) An LGC-GDAN is proposed for remote sensing scene images SR. Given the complex spatial distribution of land cover elements caused by over-head imaging angle, the context-aware network and the corresponding edge-assisted network are designed. Considering the smaller ground objects and more complex scenes caused by long-range imaging distances, dual-region level discriminators were designed.

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Moreover, two task-driven loss functions are developed to generate diverse SR image for further scene classification task.

2) To capture the complex spatial distribution of RSIs caused by over-head imaging angle, the GAS mechanism uses high-pass filtering and low-pass filtering to extract long-distance low and high-frequency information, while LAS mechanism uses local aggregation to extract short-distance low and high-frequency information. The edge-assisted branch adopts an encoder-decoder structure to learn clear and accurate object contours.

3) To cope with the challenges of small ground objects and complex scenes caused by long-range imaging distance, dual region-level discriminators are proposed in both image and edge domains to help increase the local consistency and sharpen the details of the complex scenes. Different from traditional discriminators, our discriminator is used to obtain the real/fake label of each local region in the remote-sensing scene images. The LGC-GDAN generator can benefit from the proposed discriminators to facilitate low and high-frequency information restoration.

The rest of this article is organized as follows. In Section II, the related works are briefly reviewed. In Section III, a detailed description of our method is presented. Section IV gives the experimental settings and results. The ablation experiments and scene classification results are discussed in Section IV. Finally, the conclusions are drawn in Section V.

II. Related Works

Deep CNNs and GAN [36], [37] have a wide range of applications in image SR. In this section, the relevant works of SR methods on natural images and RSIs are briefly reviewed.

A. SR for Natural Images

In recent years, the study based on deep learning architecture has attracted widespread attention [38]. Currently, numerous learning-based SR algorithms have been proposed and have achieved great success on natural images. Dong et al. [39] proposed a pioneering SRCNN, which employed a three-layer convolution structure to learn the mapping between LR images and HR images. Kim et al. [40] designed a very deep SR (VDSR) network with 20 convolutional layers and adopted residual learning in the network to overcome the difficulty of optimizing the deeper network. Zhang et al. [41] developed a residual dense network (RDN) that learns the hierarchical features from all previous layers via densely connected layers and global residual connections. Later, Haris et al. [42] designed dense deep back-projection networks (D-DBPNs) based on the idea of iterative up-sampling and down-sampling. Inspired by the idea of squeeze-and-excitation block [43], Zhang et al. [44] developed a very deep residual network named RCAN. The feature learning ability is enhanced by incorporating the channel attention module into the residual in residual dense blocks. Recently, Dai et al. [45] designed a second-order attention network (SAN) by exploring second-order statistics of features to adaptively rescale the channel-wise features. However, the methods mentioned above aim to minimize the difference in pixel values between the SR images and HR images, ignoring human perceptions [46], [47].

To generate realistic images, Ledig et al. [48] proposed super-resolution GAN (SRGAN) to recover photorealistic textures for natural images. The perceptual loss is first proposed in the SRGAN to enhance the visual quality by minimizing the error in a feature space instead of pixel space. The aim of GAN is to address the adversarial min-max problem, the parameters of the discriminator network D_{θ_D}, and generator network G_{θ_G} need to be updated as follows:

$$\min_{\theta_G}\max_{\theta_D}\mathbb{E}_{I^{\mathrm{HR}}\sim p_{\mathrm{data}}(I^{\mathrm{HR}})}\left[\log D_{\theta_D}\left(I^{\mathrm{HR}}\right)\right] + \mathbb{E}_{I^{\mathrm{LR}}\sim p_G(I^{\mathrm{LR}})}\left[\log\left(1-D_{\theta_D}\left(G_{\theta_G}\left(I^{\mathrm{LR}}\right)\right)\right)\right]. \quad (1)$$

To obtain the multihierarchical deep features, RRDBs [49] are proposed in enhanced SR GANs (ESRGANs). Subsequently, Ma et al. [37] exploit the gradient maps of the images to guide high-frequency details restoration and enhance the SR fidelity by reducing structural distortions. Recently, researchers have been developed some SR methods for promoting the resolution of different types of RSIs, e.g., optical images, radar images, and hyperspectral images [50].

B. SR for RSIs

In the field of remote sensing, SISR algorithms have attracted a great deal of attention recently, as high-quality RSIs are essential for subsequent tasks, such as object recognition and classification [35]. Recently, with the popularity of neural networks, lots of efforts have made to build robust SR architectures and pursue high-quality HR RSIs via learning a mapping relationship between LR and HR image pairs. Motivated by the success of CNNs in natural image SR, Lei et al. [31] proposed an LGCNet based on deep CNNs for learning multilevel representations of RSIs. Inspired by backprojection network [42], Pan et al. [51] proposed a residual dense backprojection network (RDBPN) including several residual backprojection blocks, which can restore the detailed structure of LR RSIs in the median and large-scale factors. To extract salient edge features, Jiang et al. [27] proposed the EEGAN to generate images with high credibility and clear contents. Considering the multiscale characteristics of ground objects in RSIs, the multiscale attention network (MSAN) [32] is proposed for learning powerful feature representations to compensate for the missing details of images. Furthermore, the scene-adaptive strategy is proposed to accurately describe the structural characteristics of different scenes [52].

Since Sentinel-2 imagery is freely available, it has been increasingly used for SR tasks. However, there are no corresponding HR-RSIs existing for Sentinel-2 images, which cause obstacles to the training of SR models [53]. Salgueiro Romero et al. [54] proposed using the artificially generated LR-HR image pairs from WorldView data to pretrain the model, and then fine-tuned the model using the WorldView-Sentinel pairs. To address the problem of image pairs,

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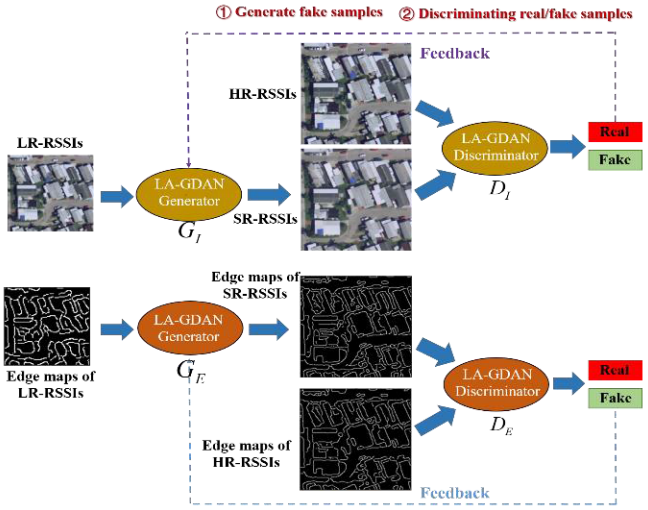

Fig. 2. Overall architecture of the LGC-GDAN. G_I is an image generator and G_E is an edge generator. D_I is the image discriminator and D_E is the edge discriminator.

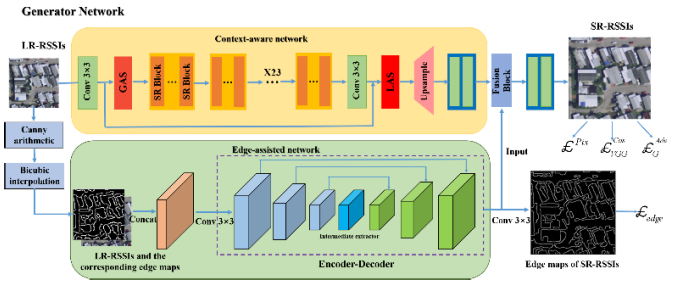

Fig. 3. Architecture of context-aware and edge-assisted network. LR-RSSIs are processed using the canny algorithm to generate edge maps. Bicubic interpolation is applied to match the spatial size of LR-RSSIs and their corresponding edge maps with HR-RSSIs. The intermediate extractor uses a residual network with dilated convolutional layers for multilevel feature extraction.

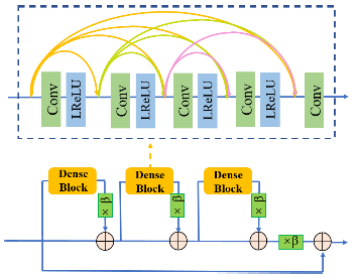

Fig. 4. Classical basic blocks RRDBs. β is the residual scaling factor.

Wang et al. [55] designed two generative CNNs for unsupervised pixel-to-pixel training. To the best of our knowledge, CNN-based methods usually produce unnatural artifacts, i.e., geometric distortions when recovering details [56]. Therefore, GAN is used as the basis of our work, and some improvements are made to generate visually pleasing SR images.

III. Proposed Method

The architecture of the proposed LGC-GDAN is shown in Fig. 2. As can be observed from Fig. 2, the LGG-GDAN network consists of dual-path generators, i.e., the context-aware network and edge-assisted network, and dual region-level discriminators, i.e., image discriminator and edge discriminator, respectively. First, the LR RSIs and edge maps are processed through the two generators to produce HR-RSIs (SR-RSSIs) and corresponding HR edge maps. Then, dual region-level discriminator networks are designed to discriminate the authenticity of the generated samples and the real samples. Furthermore, two task-driven loss function are designed to produce diverse SR results for scene classification. The components and loss functions of LGC-GDAN are elaborated in more detail in Sections III-A–III-C.

A. Context-Aware and Edge-Assisted Generator Network

The generator of LGC-GDAN is an end-to-end network, as shown in Fig. 3. It is composed of a context-aware network and an edge-assisted network. The backbone of the generator network is a residual network with 23 recursive RRDBs. Since RRDBs are the classical basic blocks of SR models, which consist of multiple weighted dense blocks and jump connections, here RRDBs are described in detail. As shown in Fig. 4, all batch normalization (BN) layers are removed, only including the convolution layers, leaky rectified linear unit (LeakyReLU), and dense connections. To avoid the problem of vanishing gradients, the residual scaling β is introduced to RRDBs to ensure stable training of the model. Several RRDBs are stacked to learn multilevel features and fine-grained image structures. To make the output images have the same spatial size as HR-RSSIs, two pixels shuffler modules are added after the feature extraction network.

RRDBs have the ability to learn the local information from the RSIs, but it is difficult to express the global information. To effectively exploit the abundant low and high-frequency in LR-RSSIs, the GAS module is designed and plugged into the head of the context-aware network to encode spatial features at the global scale. The structure of GAS is shown in Fig. 5. First, four 1×1 convolutions are employed to process the input feature maps and produce four different types of feature maps. The obtained feature maps are converted into 2-D arrays through the flattening operation to facilitate the calculation of feature correlations between each pixel. The generated feature maps belong to the original frequency features, which are represented by F_1, F_2, F_3, and F_4. The low-frequency features LF and the high-frequency features HF can be produced through the average pooling layer and the maximum pooling layer, respectively. The correlation of each pixel is calculated using F_2 and F_3 and the correlation coefficients are normalized by the softmax function to obtain the regularized correlation coefficient RCC, the mathematical operation is shown in (2). Then, HF are multiplied by the regularized correlation coefficients to extract the correlation between the low-frequency features LCC and the high-frequency features HCC, respectively, as shown in (3) and (4). Finally, the reweighted feature maps are cascaded to obtain the final output

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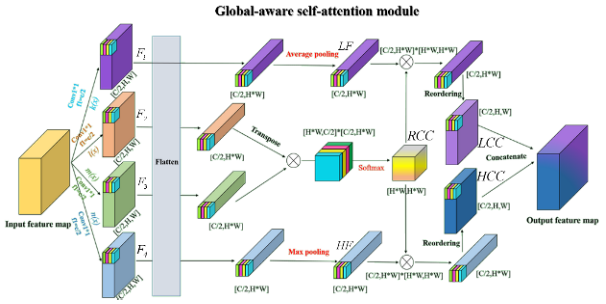

Fig. 5. Structure of GAS module. F_1, F_2, F_3, and F_4 is the original frequency features. LF is low-frequency features. HF is the high-frequency features. RCC is regularized correlation coefficient, LCC is the correlation of low-frequency features, and HCC is high-frequency feature dependencies.

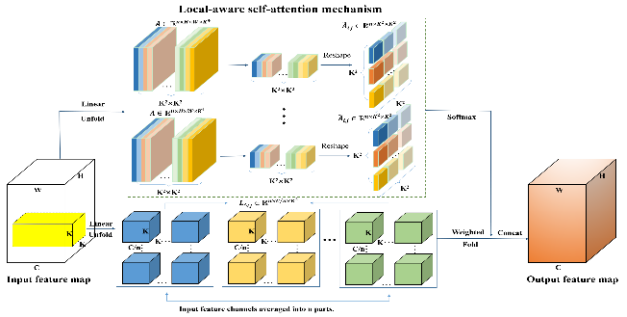

Fig. 6. Structure of LAS module. $A \in \mathbb{R}^{n\times H\times W\times K^4}$ is the attention weights; the pixel (i, j) attention weights is $\widetilde{A}_{i,j} \in \mathbb{R}^{n\times K^2\times K^2}$; the value within the local window is $L_{\Delta_{i,j}} \in \mathbb{R}^{n\times C/n\times K^2}$.

feature maps

$$\mathrm{RCC} = \mathrm{soft\,max}\left(\mathrm{Flatten}(l(X))\mathrm{Flatten}(m(X))^T\right) \tag{2}$$

$$\mathrm{LCC} = \mathrm{Reshape}(\mathrm{Average\ \ pooling}(\mathrm{Flatten}(k(X))) \cdot \mathrm{RCC}) \tag{3}$$

$$\mathrm{HCC} = \mathrm{Reshape}(\mathrm{Max\ \ \ pooling}(\mathrm{Flatten}(n(X))) \cdot \mathrm{RCC}). \tag{4}$$

Here, X refers to input feature maps, $k(x), l(x), m(x)$, and $n(x)$ respectively denotes four different convolution functions. RCC denotes regularized correlation coefficient, LCC represents the correlation of low-frequency features, and HCC indicates high-frequency feature dependencies.

To restore the fine details of the LR RSIs, the LAS mechanism is designed to model the local spatial dependencies and focus on more informative features, as shown in Fig. 6. Unlike traditional attention mechanisms that use global pooling and average pooling operations, the proposed LAS adopts unfold and fold operations to generate attention weights and pays more attention to the spatial correlation at a local scale. LAS employs a multihead attention mechanism to simultaneously model the local spatial features and spectral features [57]. Therefore, the channels of the input feature map are averaged into n parts. For each pixel (i, j), LAS is able to calculate the interdependencies across all the neighbors within a local window of size $K \times K$. Then two linear layers are employed to generate attention weights $A \in \mathbb{R}^{n\times H\times W\times K^4}$ and the value within the local window $L_{\Delta_{i,j}} \in \mathbb{R}^{n\times C/n\times K^2}$, respectively. Attention weight at location (i, j) is reshaped to $\widetilde{A}_{i,j} \in \mathbb{R}^{n\times K^2\times K^2}$ and regularized by Softmax function, as shown in (5). The obtained results are similar to the normalized covariance matrix, which is used to characterize the correlations of spatial-wise features

$$\widetilde{A}_{i,j} = \mathrm{Reshape}(\mathrm{Linear}(\mathrm{Unfold}(X))). \tag{5}$$

Then a weighted average operation is used for each local feature to encode the region-level information. Finally, the folding operation is employed to generate the spatial attention map, as shown in (6). Since the ground objects in RSIs have strong spatial agglomeration, the local spatial dependencies are essential for aggregating homogeneous information and reconstructing perceptually pleasing texture

$$O_{\Delta_{i,j}} = \mathrm{Concat}\left(\mathrm{Fold}\left(\mathrm{MatMul}\left(\mathrm{Softmax}\left(\widetilde{A}_{i,j}\right), L_{\Delta_{i,j}}\right)\right)\right). \tag{6}$$

The edge-assist branch is used to improve the recovery of high-frequency details and consists of an encoder, an intermediate extractor, and a decoder. Since RSIs inherently contain abundant edge information, they are susceptible to noise interference. The Canny arithmetic is a sample and effective method which uses Gaussian filtering to reduce noise interference [58], [59]. At the beginning of this branch, the canny arithmetic is used to process LR-RSSIs and generate corresponding edge maps. The bicubic interpolation is introduced to resize the LR-RSSIs and corresponding edge maps to be the same spatial size as HR-RSSIs. The upsampled images and edge maps are concatenated and fed into the fully convolutional network to learn multiscale hierarchical features. The skip-connected layers are used to fuse the multilevel spatial features in the encoder and abstract semantic features in the decoder. The intermediate extractor is a residual network consisting of several dilated convolutional layers to extract multilevel representations. The decoder adopts the transposed convolutional layers to recover the spatial size of the feature maps. The edge maps of SR-RSSIs can be produced and optimized through backward propagation. At the end of the generator network, the fusion block is designed to merge the feature maps generated from two branches and output the final SR-RSSIs.

B. Dual Region-Level Discriminator Networks

Inspired by the work in [28], dual discriminators D_I and D_C are designed for the image and edge domains to discriminate real HR-RSSIs from generated SR samples. With adversarial training in both image and edge domains, the generator can learn the fine appearance and avoid geometric distortions. However, the quality of the SR images largely depends on discriminator performance. When the discriminator is weak, the generator easily "fools" the discriminator. When the discriminator is strong, the recovered image is not realistic enough and is easy to be distinguished by the discriminator. Thus, the discriminator plays a vital role to improve the stability of GAN training.

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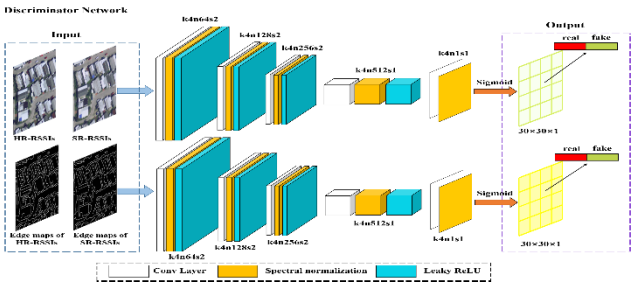

Fig. 7. Architecture of the dual region-level discriminator networks in image and edge domains.

The architecture of the dual region-level discriminator networks is illustrated in Fig. 7. It is composed of several convolutional layers, spectral normalization (SN), LeakyReLU, and sigmoid activation functions. There are four convolutional layers with an increasing number of 4×4 filter kernels, increasing by a factor of 2 from 64 to 512 kernels. This design avoids the use of maximal pooling throughout the network and focuses more on the identification of authentic samples at the regional level. The SN is added after each convolutional layer to guarantee stable training of the LGC-GDAN. Compared with other regularization techniques, such as weight normalization and gradient penalty, the SN technique can control gradient vanishing throughout the training and improve the sample quality of GANs. The LeakyReLU is a nonlinear activation function and is plugged after the SN to nonlinearly transform the feature maps into more discriminative representations. The sigmoid activation function is used to output the labels.

Since the RSIs have the characteristic of small ground objects and complex scenes caused by imaging distance, it is unreasonable to distinguish the real/fake of the input data based on the spatial information of the entire image. Different from traditional discriminator, a region-level discriminator is developed to distinguish the real/fake of local regions. In the LGC-GDAN, the output of the discriminator is no longer real/fake labels of the entire image, but the labels of each local grid region.

C. Multitype Joint Loss Functions

As for most SR methods, they optimize the networks by a commonly used pixel-wise loss or pixel-wise mean square error (MSE) loss, which are beneficial for SR tasks measured by peak-signal-to-noise ratio (PSNR). They are defined as

$$\mathcal{L}^{\mathrm{Pix}_I} = \frac{1}{r^2 WH} \sum_{x=1}^{rW} \sum_{y=1}^{rH} \left\| \left(G\left(I^{\mathrm{LR}}\right)\right)_{x,y} - \left(I^{\mathrm{HR}}\right)_{x,y} \right\|_1 \tag{7}$$

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{r^2 WH} \sum_{x=1}^{rW} \sum_{y=1}^{rH} \left(\left(G\left(I^{LR}\right)\right)_{x,y} - \left(I^{\mathrm{HR}}\right)_{x,y} \right)^2. \tag{8}$$

Since MSE loss can accelerate convergence and achieve higher PSNR, it is introduced to the overall loss function of LGC-GDAN to reduce the pixel value difference between recovered images and ground truths. However, many studies have shown that high-frequency information will be ignored and produce blurry results when over-relied on pixel-wise loss.

To tackle this problem, perceptual loss has been proposed to improve the perceptual quality of the recovered images. This loss is used to calculate the Euclidean distances between the feature maps, which are semantic information generated by the pretrained visual geometry group (VGG) network. Due to the rich semantic information contained in deep features, the output feature of layer 12 of the VGG model is selected for perceptual loss computation. The perceptual loss is defined as follows:

$$\mathcal{L}^{\mathrm{Per}}_{\mathrm{VGG}} = \frac{1}{r^2 W_l H_l} \sum_{x=1}^{rW_l} \sum_{y=1}^{rH_l} \| \phi^l\left(I^{\mathrm{HR}}\right)_{x,y} - \phi^l\left(G\left(I^{LR}\right)\right)_{x,y} \|_1 \tag{9}$$

where l represents the feature layer of the VGG network, $\phi^l(\cdot)$ is the feature map in the l layer, W_l and H_l are the width and height of the feature map in the l layer.

However, the perceptual loss is still utilized to calculate the difference in pixel value between the feature maps. To enhance the texture similarity between HR-RSSIs and SR-RSSIs, the semantic similarity loss function is designed to calculate the similarity of each pixel in the feature maps. The semantic similarity loss is developed based on cosine loss. The cosine distance between two features is minimized as follows:

$$\mathcal{L}^{\mathrm{Cos}}_{\mathrm{VGG}} = 1 - \frac{1}{r^2 W_l H_l} \sum_{x=1}^{rW_l} \sum_{y=1}^{rH_l} \frac{\phi_l^{(x,y)}\left(I^{\mathrm{HR}}\right) \cdot \phi_l^{(x,y)}\left(G\left(I^{\mathrm{LR}}\right)\right)}{\| \phi_l^{(x,y)}\left(I^{\mathrm{HR}}\right) \| \| \phi_l^{(x,y)}\left(G\left(I^{\mathrm{LR}}\right)\right) \|} \tag{10}$$

where $\phi_l^{(x,y)}$ refers to the pixel of the feature map in the l layer at the location (x, y).

It can be seen that the semantic similarity loss function aims to compute the similarity of high-level feature maps instead of pixel value difference. To guide the generator network to produce realistic results, the adversarial learning strategy is adopted. The adversarial losses are defined as follows:

$$\mathcal{L}_G^{\mathrm{Adv}} = -\mathbb{E}_{I^{\mathrm{LR}} \sim p_{\mathrm{LR}}}\left(\log\left(D\left(G\left(I^{\mathrm{LR}}\right)\right)\right)\right) \tag{11}$$

$$\mathcal{L}_{D_I}^{\mathrm{Adv}} = -\mathbb{E}_{I^{\mathrm{HR}} \sim p_{\mathrm{HR}}}\left[\log\left(D_I\left(I^{\mathrm{HR}}\right)\right)\right] - \mathbb{E}_{I^{\mathrm{LR}} \sim p_{\mathrm{LR}}}\left(\log\left(1 - D_I\left(G\left(I^{\mathrm{LR}}\right)\right)\right)\right) \tag{12}$$

$$\mathcal{L}_{D_C}^{\mathrm{Adv}} = -\mathbb{E}_{I^{\mathrm{HR}} \sim p_{\mathrm{HR}}}\left[\log\left(D_C\left(C^{\mathrm{HR}}\right)\right)\right] - \mathbb{E}_{I^{\mathrm{LR}} \sim p_{\mathrm{LR}}}\left(\log\left(1 - D_C\left(G\left(C^{\mathrm{LR}}\right)\right)\right)\right). \tag{13}$$

The existing SR models only consider the image space constraint but neglect the semantically structural information provided by the edge space. To avoid over-smoothing or over-sharpening of SR-RSSIs, the edge loss is introduced into the overall loss function of the generator. The edge loss is formulated by diminishing the Euclidean distance between the edge maps of HR-RSSIs and SR-RSSIs, as described below

$$\mathcal{L}_{\mathrm{edge}} = \frac{1}{r^2 WH} \sum_{x=1}^{rW} \sum_{y=1}^{rH} \| \mathrm{Edge}\left(I^{\mathrm{SR}}\right) - \mathrm{Edge}\left(I^{\mathrm{HR}}\right) \|_1. \tag{14}$$

In summary, dual discriminator networks D_I and D_C are optimized by $\mathcal{L}_{D_I}^{\mathrm{Adv}}$ and $\mathcal{L}_{D_C}^{\mathrm{Adv}}$. The generator

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network is optimized through the joint loss function, which is defined as follows:

$$\mathcal{L}_G^{\text{Total}} = \lambda_1 \mathcal{L}_{\text{MSE}} + \lambda_2 \mathrm{L}_{\text{VGG}}^{\text{Cos}} + \lambda_3 \mathcal{L}_G^{\text{Adv}} + \lambda_4 \mathcal{L}_{\text{edge}}. \quad (15)$$

In our experiments, the parameters are set to $\lambda_1 = 1$, $\lambda_2 = 0.01$, $\lambda_3 = 0.001$, and $\lambda_4 = 1$. In this way, the recovered images have high PSNR and structural similarity index measure (SSIM) but they are not realistic enough. To produce the sharpened SR-RSSIs, another joint loss function is designed based on perceptual loss, as follows:

$$\mathcal{L}_G = \mu_1 \mathcal{L}_{\text{MSE}} + \mu_2 \mathcal{L}_{\text{VGG}}^{\text{Per}} + \mu_3 \mathcal{L}_G^{\text{Adv}}. \quad (16)$$

The optimization procedure of LGC-GDAN is described by pseudocode, as shown in Algorithm 1. Here, $P_I = \{I_k^{\text{SR}}, I_k^{\text{HR}}\}_{k=1}^{k=k_B}$ refers to the set consisting of SR and HR RSI pairs. $N(\cdot)$ denote discriminator of the LGC-GDAN without sigmoid function. $e^{-N(P_I)} > 1$ and $e^{-N(P_I)} < 1$ are conditional judgments. In the sigmoid function, when $e^{-N(P_I)} > 1$, then $(1/1+e^{-N(P_I)}) < 0.5$, the judgment result of the discriminator is zero; when $e^{-N(P_I)} < 1$, then $(1/1 + e^{-N(P_I)}) > 0.5$, the judgment result of the discriminator is one, as shown in the equation

$$D_I(P_I) = \begin{cases} 0, & \text{if } e^{-N(P_I)} > 1 \\ 1, & \text{if } e^{-N(P_I)} < 1. \end{cases}$$

IV. Experiments

In this section, we first introduce the datasets used for the experiments and experimental details. Then, the reconstruction performance of our method and other SR models was compared. Furthermore, extensive ablation experiments were conducted to evaluate the effectiveness of our design choice. Finally, the feasibility of SR-RSSIs in the scene classification task was discussed.

A. Experimental Datasets

1) AID [60] is a large-scale collection of RSIs including more than 10 000 aerial scene images. The sample images are collected from Google Earth imagery, they have high intraclass diversity and low interclass dissimilarity. AID is made up of 30 land-use classes, including airport, beach, commercial, residential, meadow, river, etc. Since AID is a multisource remote sensing data and contains a large number of feature types, it is very suitable as a training dataset for RSI SR [28].

2) RSSCN7 dataset [61] is also collected from Google Earth, which contains 2800 aerial scene images divided into seven scene categories. There are 400 images with a size of 400 × 400 pixels for each class. It is worth noting that the images of each category are obtained under different scales and imaging conditions, which improves the applicability of the dataset.

3) UC-Merced dataset [62] is composed of 21 land-use classes, and the images are selected from aerial orthoimagery with a pixel resolution of 1 ft. Moreover, these images cover many representative regions of the United States. There are 2100 images in total, and each class has 100 samples with a fixed spatial size of 256 × 256.

Algorithm 1 Optimization Procedure of LGC-GDAN

Input: LR-RSSIs: I^{LR},
HR-RSSIs: I^{HR},
Output: The parameters of generator and discriminator networks: $\theta = \{\theta_G, \theta_{D_I}, \theta_{D_C}\}$
Initialization: $\theta_G^1, \ldots, \theta_G^M$, $\theta_{D_I}^1, \ldots, \theta_{D_I}^M$, $\theta_{D_C}^1, \ldots, \theta_{D_C}^M$
while training loss has not converged **do**
(1) Update G
Draw k_B samples from I^{LR} and I^{HR} image pairs, $\{I_k^{LR}, I_k^{HR}\}_{k=1}^{k=k_B}$
for $k = 1,\ldots,k_B$ **do**
$C_k^{LR} \leftarrow Canny(I_k^{LR})$
$I_k^{SR}, I_k^{edge} \leftarrow G(I_k^{LR}, C_k^{LR})$
end for
Update θ_G^i by minimizing $\mathcal{L}_G^{Total}$ in Equation (15) through stochastic gradient descent and weight decay:

$$\theta_G^i \leftarrow \left(1 - \frac{\eta\lambda}{n}\right)\theta_G^i - \eta \frac{\partial \mathcal{L}_G^{Total}}{\partial \theta_G^i}$$

(2) Update D_I and D_C
Feature extraction network of discriminator: $N(\cdot)$
Draw k_B samples from I^{SR} and corresponding I^{HR}, $P_I = \{I_k^{SR}, I_k^{HR}\}_{k=1}^{k=k_B}$, $P_C = \{C_k^{SR}, C_k^{HR}\}_{k=1}^{k=k_B}$
for $k = 1,\ldots, k_B$ **do**

$$D_I(P_I) = \begin{cases} 0, & if \quad e^{-N(P_I)} > 1 \\ 1, & if \quad e^{-N(P_I)} < 1 \end{cases}$$

end for
for $k = 1,\ldots,k_B$ **do**

$$D_C(P_C) = \begin{cases} 0, & if \quad e^{-N(P_C)} > 1 \\ 1, & if \quad e^{-N(P_C)} < 1 \end{cases}$$

end for
Update $\theta_{D_I}^i$ and $\theta_{D_C}^i$ by minimizing the sum of $\mathcal{L}_{D_I}^{Adv}$ and $\mathcal{L}_{D_C}^{Adv}$ in Equation (12), (13) through stochastic gradient descent:

$$\theta_{D_I}^i \leftarrow \theta_{D_I}^i - \eta \frac{\partial(\mathcal{L}_{D_I}^{Adv} + \mathcal{L}_{D_C}^{Adv})}{2\partial\theta_{D_I}^i}$$

$$\theta_{D_C}^i \leftarrow \theta_{D_C}^i - \eta \frac{\partial(\mathcal{L}_{D_I}^{Adv} + \mathcal{L}_{D_C}^{Adv})}{2\partial\theta_{D_C}^i}$$

end while

4) WHU-RS19 dataset [63] is a set of satellite images from Google Earth. The images have a fixed spatial size of 600 × 600 pixels with various pixel resolutions. In addition, the dataset consists of a total of 1005 images labeled with 19 classes of physical scenes. The dataset has about 50 sample images per category, which is a relatively small number than other datasets.

B. Implementation Details

To meet the data requirement of model training, LR-RSSIs are produced by downsampling HR-RSSIs with a bilinear

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kernel. The spatial size of LR-RSSIs is resized to 64×64 and corresponding HR-RSSIs are resized to 256×256. We mainly focus on the upscaling factor of 4, which is usually the most challenging scenario in image SR. The experiments are trained on the AID and tested on the WHU-RS19, UCM-21, and RSSCN7 datasets to verify the performance of our proposed LGC-GDAN. It is important to emphasize that we did not augment the training dataset. The PSNR and SSIM are used to quantitatively evaluate the quality of the SR solutions. PSNR is the most common index to estimate absolute errors between HR-RSSIs and SR-RSSIs. SSIM is a perception-based metric that perceives the difference between images in structural information. Nevertheless, SR-RSSIs with high PSNR and SSIM scores may be overly smooth and lack realistic visual perception. Therefore, apart from the PSNR and SSIM, the learned perceptual image patch similarity (LPIPS) is also used as an evaluation metric in image SR. The LPIPS is employed to measure image similarities using Euclidean distance in the feature space of the pretrained AlexNet model. On the contrary to PSNR and SSIM, lower LPIPS denote better SR results. With regard to model training, the Adam optimizer is used for optimization with the parameters of $\delta_1 = 0.9$, $\delta_2 = 0.999$, and $\varepsilon = 1 \times 10^{-8}$. The learning rate of both generator and discriminators is set to 0.0001, and the epoch number is ten. The number of channels of the RRDBs is set to 64 and 32. The proposed LGC-GDAN is sensitive to the choice of the regularization parameter λ_1, λ_2, λ_3, and λ_4 in the overall joint loss function. In this article, the regularization parameters are set to 1, 0.01, 0.001, and 1, respectively. To produce visually favorable results, the regularization parameter β_1, β_2, β_3 in (16) are set to 0.1, 1, and 0.001, respectively. All experiments are implemented by PyTorch framework and evaluated on a workstation equipped with four NVIDIA GeForce RTX 2080Ti GPU and a workstation with one RTX 5000 GPU.

C. Quantitative and Qualitative Comparison of Different Methods on Different Datasets

In this section, the experimental results of our method and state-of-the-art SISR methods on the three remote sensing scene datasets are compared. There are five CNN-based SISR methods (i.e., SRCNN [39], VDSR [40], LGCNet [31], D-DBPN [42], and RCAN [44]), two GAN-based SISR methods (i.e., SRGAN [48] and EEGAN [27]), and a transformer-based method (i.e., HAT [67]). Especially, LGCNet and EEGAN are specially designed for remote sensing SR. All these methods are trained on the AID for a fair comparison. Note that all images in the training and test datasets are resized to 256×256 pixels as HR-RSSIs and 64×64 pixels as LR-RSSIs.

To demonstrate the influence of different loss functions on the results, experiments are conducted on two types of loss functions, namely LGC-GDAN and LGC-GDAN+. The former adopts sematic similarity loss, edge loss, GAN loss, and pixel loss. The latter employs perceptual loss, GAN loss, and pixel loss. We quantitatively evaluated the SR results using three metrics, including PSNR, SSIM, and LPIPS. To analyze the reconstruction performance of each class, the PSNR, SSIM, and LPIPS of each class are given in detail. In each row, the best results are highlighted in bold. The SR results of the existing methods and our method on the three datasets are listed in Tables I–III. Due to the limitation of the space, we only present the results of the difficult-to-reconstruct category.

From these tables on scale $\times 4$, we can see that LGC-GDAN achieves the highest value in the metric of PSNR and SSIM, while LGC-GDAN+ exhibits better accuracy in the metric of LPIPS. Generally speaking, CNN-based methods have better PSNR and SSIM than GAN-based methods because they focus on recovering the spatial structure of the LR-RSSIs. However, the results generated by the CNN-based methods lack a realistic visual appearance, which causes worse LPIPS. In contrast, GAN-based methods can obtain better LPIPS. It is because the adversarial loss and similarity loss is able to encourage the generator to produce visually favorable results. Besides, as the number of model layers deepens, better-quality images can be generated. It can be seen from the results of SRGAN and EEGAN that residual structure plays a key role in remote sensing SR, which can make full use of the abundant low-frequency information of LR-RSSIs to prevent the gradient from vanishing and improve PSNR and SSIM. It can be found from the tables that RCAN has higher scores on PSNR and SSIM than other SISR methods. It demonstrates that the attention mechanism is able to achieve interchannel information interaction and focus on fine structure restoration. The tables illustrate that the HAT method performs poorly compared to LGC-GDAN. The experimental results showed that the method that performs well for natural images is still unsatisfactory in the SR task of RSIs. It sufficiently indicates that our method can more adequately capture the local-global high-low frequency information and rich detail information of RSIs. Compared with GAN-based methods, LGC-GDAN is not only very competitive in the metric of PSNR and SSIM but also performs well on LPIPS. The reason is that LGC-GDAN can learn abundant texture similarity and color similarity through GAS and LAS to produce high-quality SR-RSSIs. Moreover, LGC-GDAN+ achieves the best values on the LPIPS because it focuses more on the restoration of high-frequency details to obtain more realistic results.

The numerical analysis from the tables finds that there are huge differences in the reconstruction results of different scene categories. It shows that the complexity and diversity of the land-cover types in the RSIs bring a huge challenge for SR tasks. As shown in Table II, PSNRs of the Grass, Field, and RiverLake scenes are relatively high, while PSNRs of the Parking, Industry, and Resident scenes are relatively lower. This is because scene images with high PSNR contain few and simple land cover types, so less complex texture and structure information need to be recovered. In this article, the spatial context, edge, and structure information are integrated to reconstruct both low and high-frequency signals of the LR-RSSIs.

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TABLE I

SR RESULTS OF SRCNN [39], VDSR [40], LGCNET [31], SRGAN [48], D-DBPN [42], EEGAN [27], RCAN [44], HAT [67], AND PROPOSED METHOD ON THE UC-MERCED [62] DATASET

UC-Merced	Metrics	SRCNN	VDSR	LGCNet	SRGAN	D-DBPN	EEGAN	RCAN	HAT	LGC-GDAN	LGC-GDAN+
Forest	PSNR	25.8847	26.2591	26.2229	25.4194	26.3591	26.1786	26.5108	26.2428	**26.7269**	25.6650
	SSIM	0.5875	0.6096	0.6081	0.6233	0.6248	0.6103	**0.6288**	0.5362	0.6248	0.5551
	LPIPS	0.3026	0.2672	0.2761	0.2585	0.2513	0.2499	0.2685	0.3165	0.2606	**0.1610**
Freeway	PSNR	25.5258	26.1096	25.9911	26.3097	27.3183	26.5877	27.4226	25.5647	**27.9265**	27.0495
	SSIM	0.6936	0.7118	0.7088	0.7340	0.7591	0.7281	**0.7656**	0.6562	0.7565	0.7199
	LPIPS	0.2349	0.1898	0.2100	0.1545	0.1353	0.1667	0.1468	0.2211	0.1445	**0.1045**
Tennis court	PSNR	25.9859	26.5720	26.4550	26.9193	27.5378	26.7302	27.7046	25.5447	**28.1877**	27.2086
	SSIM	0.7145	0.7312	0.7274	0.7582	0.7761	0.7360	**0.7842**	0.6652	0.7792	0.7390
	LPIPS	0.2535	0.2000	0.2216	0.1617	0.1431	0.1798	0.1536	0.2435	0.1472	**0.0965**
Sparse residential	PSNR	25.7934	26.2656	26.1937	26.3350	26.9429	26.3715	26.9425	25.5938	**27.5307**	26.5640
	SSIM	0.6648	0.6786	0.6768	0.6998	0.7042	0.6790	**0.7086**	0.6166	0.7050	0.6581
	LPIPS	0.2786	0.2346	0.2522	0.2066	0.2090	0.2205	0.2253	0.2822	0.2125	**0.1379**
Airplane	PSNR	24.9824	25.7853	25.5417	26.0953	27.0347	26.2790	27.0302	24.3039	**27.6166**	26.8802
	SSIM	0.7125	0.7332	0.7249	0.7522	0.7711	0.7437	**0.7764**	0.7003	0.7730	0.7393
	LPIPS	0.2479	0.1997	0.2196	0.1611	0.1453	0.1790	0.1583	0.2386	0.1470	**0.1008**
Baseball diamond	PSNR	29.1402	29.6619	29.5968	29.1210	30.3335	29.8134	30.3212	29.0544	**31.5783**	30.5330
	SSIM	0.7731	0.7822	0.7808	0.7909	0.7995	0.7753	**0.8054**	0.7524	0.8008	0.7697
	LPIPS	0.2570	0.2204	0.2356	0.1857	0.1808	0.1929	0.2030	0.2687	0.1801	**0.1319**
Buildings	PSNR	23.0790	23.9044	23.6370	24.2591	25.1516	24.3063	25.2432	22.2329	**25.6528**	24.8425
	SSIM	0.7016	0.7336	0.7203	0.7575	0.7881	0.7477	**0.7924**	0.6601	0.7901	0.7573
	LPIPS	0.2209	0.1719	0.1936	0.1438	0.1163	0.1595	0.1241	0.2024	0.1164	**0.0860**
Medium residential	PSNR	23.3770	23.8990	23.7635	24.4509	24.9143	24.1844	24.9315	23.6847	**25.2864**	24.4691
	SSIM	0.6475	0.6713	0.6640	0.7070	0.7222	0.6864	**0.7250**	0.6209	0.7249	0.6824
	LPIPS	0.2826	0.2288	0.2513	0.1969	0.1808	0.2175	0.1941	0.2579	0.1821	**0.1158**
Parking lot	PSNR	20.3032	20.7952	20.6907	21.0608	21.4870	21.0194	21.5204	19.2069	**21.6836**	21.0518
	SSIM	0.6469	0.6802	0.6710	0.6975	0.7307	0.6938	0.7323	0.5485	0.7355	0.6975
	LPIPS	0.2341	0.1834	0.1996	0.1601	0.1300	0.1741	0.1437	0.2276	0.1246	**0.0942**
Overpass	PSNR	23.7353	24.3540	24.1253	24.9506	25.8052	24.8303	25.7733	23.6463	**26.0880**	25.4948
	SSIM	0.6480	0.6715	0.6631	0.7071	0.7347	0.6920	**0.7378**	0.6172	0.7298	0.6950
	LPIPS	0.2705	0.2210	0.2459	0.1794	0.1530	0.1972	0.1652	0.2457	0.1620	**0.1167**
Dense residential	PSNR	23.3720	24.1469	23.9136	24.7278	25.4466	24.5562	25.4839	22.1955	**25.8108**	24.8754
	SSIM	0.6969	0.7286	0.7168	0.7600	0.7864	0.7462	**0.7905**	0.6266	0.7883	0.7491
	LPIPS	0.2296	0.1764	0.1975	0.1483	0.1228	0.1663	0.1302	0.2146	0.1228	**0.0863**

TABLE II

SR RESULTS OF SRCNN [39], VDSR [40], LGCNET [31], SRGAN [48], D-DBPN [42], EEGAN [27], RCAN [44], HAT [67], AND PROPOSED METHOD ON THE RSSCN7 [61] DATASET

RSSCN7	Metrics	SRCNN	VDSR	LGCNet	SRGAN	D-DBPN	EEGAN	RCAN	HAT	LGC-GDAN	LGC-GDAN+
Parking	PSNR	23.6814	24.0803	23.9640	24.3238	24.7461	24.2619	24.7925	22.3731	**25.0917**	24.3657
	SSIM	0.6340	0.6546	0.6473	0.6801	0.7022	0.6686	0.7008	0.5288	**0.7029**	0.6592
	LPIPS	0.2868	0.2442	0.2628	0.2092	0.1987	0.2288	0.2086	0.2245	0.2011	**0.1180**
Resident	PSNR	22.5947	22.9921	22.8930	23.2570	23.6953	23.1632	23.7114	21.3837	**23.9132**	23.0923
	SSIM	0.6304	0.6552	0.6488	0.6803	0.7056	0.6699	0.7046	0.5038	**0.7063**	0.6518
	LPIPS	0.2907	0.2454	0.2654	0.2164	0.2066	0.2380	0.2198	0.2296	0.2076	**0.1109**
Field	PSNR	31.6632	31.9954	32.0342	30.7079	32.1188	31.6889	32.4518	30.9097	**34.0380**	32.9172
	SSIM	0.7895	0.7953	0.7937	0.7807	0.8113	0.7960	**0.8116**	0.6943	0.8108	0.7796
	LPIPS	0.2744	0.2518	0.2584	0.2434	0.2332	0.2198	0.2317	0.2532	0.2171	**0.1456**
Forest	PSNR	27.3239	27.6102	27.5719	26.7016	27.6982	27.4330	27.8576	26.3289	**28.2261**	27.0804
	SSIM	0.6056	0.6187	0.6176	0.6278	0.6382	0.6218	**0.6394**	0.4965	0.6376	0.5655
	LPIPS	0.2815	0.2558	0.2635	0.2453	0.2545	0.2332	0.2559	0.2626	0.2496	**0.1453**
River/Lake	PSNR	28.6999	29.2928	29.1564	28.7906	29.7292	29.2787	29.9615	27.7971	**30.5955**	29.4795
	SSIM	0.7983	0.8104	0.8062	0.8142	0.8320	0.8125	0.8324	0.7371	**0.8333**	0.7968
	LPIPS	0.2277	0.1889	0.2045	0.1645	0.1599	0.1703	0.1637	0.1851	0.1606	**0.0965**
Grass	PSNR	31.0846	31.4434	31.4164	29.9443	31.8195	31.4032	31.9528	31.0916	**33.2286**	32.0034
	SSIM	0.8034	0.8101	0.8091	0.7924	0.8264	0.8088	**0.8264**	0.7548	0.8259	0.7828
	LPIPS	0.2679	0.2437	0.2539	0.2308	0.2288	0.2091	0.2281	0.2517	0.2114	**0.1330**
Industry	PSNR	23.4307	23.9428	23.7598	24.3582	24.8580	24.2090	24.8563	22.4758	**25.1920**	24.4331
	SSIM	0.6555	0.6806	0.6711	0.7093	0.7341	0.6967	0.7324	0.5593	**0.7352**	0.6915
	LPIPS	0.2705	0.2231	0.2455	0.1871	0.1721	0.2098	0.1825	0.1992	0.1729	**0.1021**

In addition to the quantitative evaluate, visual comparisons between the proposed method and other methods are also performed, as shown in Figs. 8–10. From the recovered images of different methods, we can find that our proposed method is able to alleviate the blurring artifacts and recover more spatial details so that SR-RSSIs are closer to the real HR-RSSIs. Through qualitative comparisons with state-of-the-art SR methods, only our method can recover a clear and accurate pattern. Figs. 9 and 10 show that the ground objects in our recovered images have finer contours and less geometric distortion than other results. As illustrated in Fig. 8, the RCAN method exhibits blurring artifacts in the red box part, whereas the LGC-GDAN method significantly alleviates blurring artifacts. As present in Fig. 9, the other methods produce a lot of artifacts between different floors in the red box, resulting in blurred details, while the LGC-GDAN method significantly suppresses the artifacts and enhances the clarity of the details. In Fig. 10, the line widths in the red

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TABLE III

SR RESULTS OF SRCNN [39], VDSR [40], LGCNET [31], SRGAN [48], D-DBPN [42], EEGAN [27], RCAN [44], HAT [67], AND PROPOSED METHOD ON THE WHU-RS19 [63] DATASET

WHU-RS19	Metrics	SRCNN	VDSR	LGCNet	SRGAN	D-DBPN	EEGAN	RCAN	HAT	LGC-GDAN	LGC-GDAN+
Industrial	PSNR	22.1490	22.5889	22.4884	23.1124	23.8488	22.8820	23.8212	22.8544	**24.2460**	23.3256
	SSIM	0.6178	0.6479	0.6411	0.6827	0.7241	0.6680	0.7225	0.6015	**0.7304**	0.6805
	LPIPS	0.2759	0.2377	0.2596	0.2045	0.1712	0.2314	0.1889	0.1116	0.1740	**0.1017**
Parking	PSNR	21.3730	21.6937	21.6446	22.0056	22.5883	21.9100	22.5799	21.8047	**23.4313**	22.2409
	SSIM	0.6266	0.6526	0.6453	0.6831	0.7231	0.6694	0.7247	0.6484	**0.7431**	0.6959
	LPIPS	0.2846	0.2442	0.2604	0.2090	0.1703	0.2091	0.1889	0.1033	0.1659	**0.0968**
Forest	PSNR	26.0638	26.4195	26.3737	25.6965	26.5828	26.3249	26.7016	25.7127	**26.8782**	25.9418
	SSIM	0.5212	0.5480	0.5412	0.5629	0.5717	0.5475	**0.5751**	0.4983	0.5669	0.5004
	LPIPS	0.2687	0.2631	0.2679	0.2472	0.2429	0.2307	0.2580	0.1616	0.2407	**0.1611**
Residential	PSNR	20.2247	20.5887	20.5225	20.9889	21.6529	20.8187	21.6231	20.8491	**22.1240**	21.1492
	SSIM	0.5961	0.6256	0.6196	0.6508	0.7042	0.6427	0.7031	0.5713	**0.7176**	0.6571
	LPIPS	0.3046	0.2687	0.2866	0.2376	0.1979	0.2726	0.2196	0.1312	0.1969	**0.1095**
River	PSNR	25.5865	25.9528	25.9032	26.0403	26.6267	26.0327	26.6165	25.4450	**26.9421**	26.0078
	SSIM	0.6515	0.6717	0.6689	0.6954	0.7072	0.6729	**0.7121**	0.6062	0.7083	0.6558
	LPIPS	0.2607	0.2339	0.2519	0.2041	0.2034	0.2177	0.2173	0.1310	0.1985	**0.1159**
Railway Station	PSNR	23.2634	23.6115	23.5542	23.7025	24.3481	23.6988	24.4474	22.7437	**24.6979**	23.9029
	SSIM	0.5787	0.6022	0.6007	0.6344	0.6609	0.6201	0.6623	0.5094	**0.6641**	0.6114
	LPIPS	0.2800	0.2367	0.2610	0.2048	0.1787	0.2217	0.1960	**0.1008**	0.1855	0.1268
Park	PSNR	24.8357	25.2332	25.1651	25.2332	25.7743	25.2875	25.9088	24.8964	**26.1566**	25.1737
	SSIM	0.6491	0.6698	0.6674	0.6925	**0.7036**	0.6777	0.7073	0.5990	0.7073	0.6493
	LPIPS	0.2839	0.2505	0.2693	0.2311	0.2185	0.2417	0.2353	0.1438	0.2192	**0.1257**
Commercial	PSNR	21.3062	21.7223	21.6115	21.9825	22.4676	21.9161	22.4650	21.3791	**22.8693**	21.9316
	SSIM	0.6174	0.6469	0.6403	0.6626	0.6984	0.6607	0.6980	0.5491	**0.7076**	0.6507
	LPIPS	0.2885	0.2479	0.2687	0.2222	0.1995	0.2490	0.2178	0.1263	0.2001	**0.1199**
Viaduct	PSNR	21.7946	22.2361	22.0869	22.9047	23.5475	22.6987	23.5170	21.9056	**23.8781**	23.0723
	SSIM	0.5972	0.6275	0.6188	0.6706	0.7099	0.6570	0.7085	0.5415	**0.7131**	0.6642
	LPIPS	0.2696	0.2282	0.2541	0.1856	0.1569	0.2153	0.1757	**0.0979**	0.1591	0.1030
Port	PSNR	22.8129	23.3131	23.2042	23.7408	24.4039	23.5765	24.4086	22.8455	**24.9577**	23.9395
	SSIM	0.7202	0.7496	0.7429	0.7702	0.8006	0.7539	0.8039	0.7093	**0.8103**	0.7765
	LPIPS	0.2509	0.2077	0.2281	0.1722	0.1469	0.1963	0.1628	0.0954	0.1418	**0.0917**

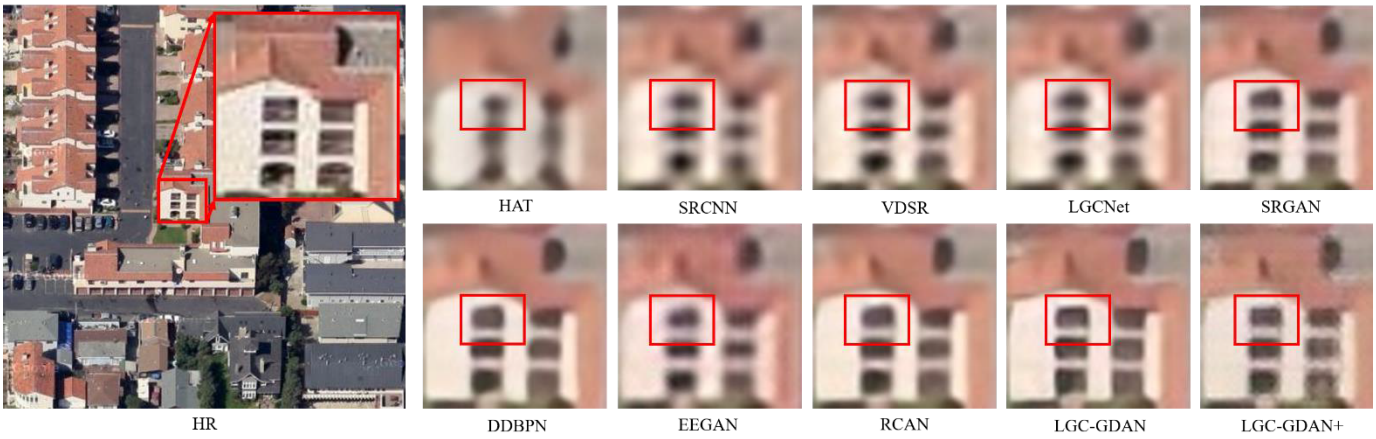

Fig. 8. Visual comparisons of some representative SR methods and our methods on $\times 4$ factor.

boxes are noticeably thinner in the RCAN method, while the LGC-GDAN and LGC-GDAN+ methods can reconstruct the line widths better. The DDBPN method also performs well, but it produces more blurring artifacts at the leftmost white line compared to the LGC-GDAN+ method. The qualitative and quantitative results declare the superiority and feasibility of the proposed LGC-GDAN for SR of RSIs.

D. Ablation Studies

In this section, a series of experiments were conducted to verify the effectiveness of each component, including edge-assisted branch, GAS module, and LAS module. The model without the adversarial learning strategy was also discussed, which was named LGC-Net.

Comparison experiments were conducted on the RSSCN7 dataset, as shown in Table IV. "Baseline" means that the residual network with several RRDBs was directly used for SR. Referring to Table IV, we can find that, the "Baseline" method has a similar performance to SRGAN. After the edge-assisted branch is equipped, the PSNR and SSIM have a slight improvement. In our experiments, adding the GAS module to LGC-GDAN aims to extract useful low and high-frequency information from the entire image. Unlike other attention mechanisms that only focus on local details, GAS pays more attention to the global spatial context. In this way, improvements of about 0.5 and 0.03 are achieved in terms of PSNR and SSIM, respectively. The results show that GAS is very beneficial for image restoration. As can be seen from Table IV, the PSNR and SSIM of LGC-Net are

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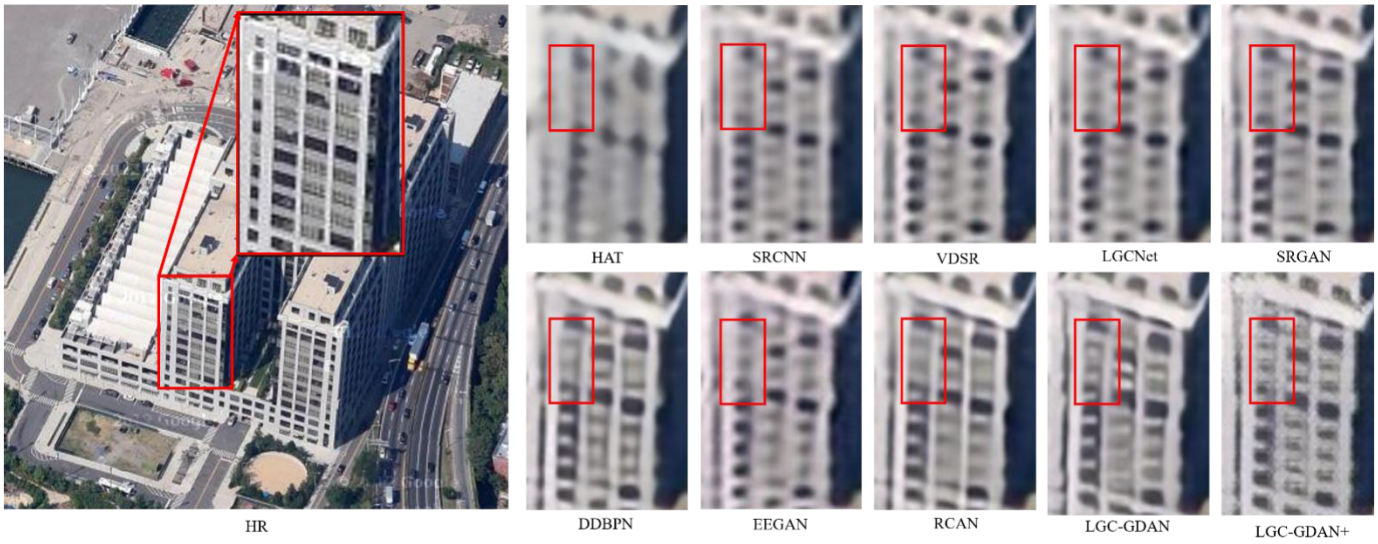

Fig. 9. SR results of remote sensing SR-based methods with a scale factor of ×4.

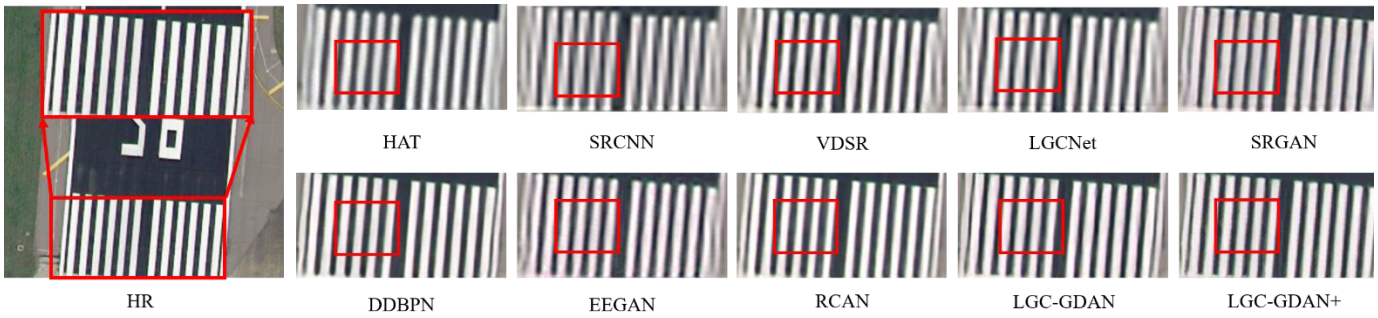

Fig. 10. Visual comparisons for 4 × SR between our model with different loss functions and other methods.

TABLE IV
RESULTS OF THE ABLATION STUDIES ON THE RSSCN7 DATASET

RSSCN7	Metric	Baseline	LGC-GDAN w/o GAS LAS	LGC-GDAN w/o LAS	LGC-Net
Parking	PSNR	24.3657	24.5826	24.9622	24.6169
	SSIM	0.6592	0.6682	0.6924	0.6677
Resident	PSNR	23.0923	23.3704	23.7811	23.2449
	SSIM	0.6518	0.6663	0.6946	0.6536
Field	PSNR	32.9172	33.3844	33.8559	33.6487
	SSIM	0.7796	0.7823	0.8096	0.8073
Forest	PSNR	27.0804	27.5096	28.1128	27.8102
	SSIM	0.5655	0.5834	0.6226	0.6074
River /Lake	PSNR	29.4795	29.8538	30.3704	29.8794
	SSIM	0.7968	0.8064	0.8195	0.8073
Grass	PSNR	32.0034	32.4143	33.1227	32.7760
	SSIM	0.7828	0.8054	0.8215	0.8192
Industry	PSNR	24.4331	24.6967	25.0426	24.5387
	SSIM	0.6915	0.7076	0.7286	0.6936

relatively worse than LGC-GDAN. It reflects the key role of the adversarial learning strategy and the proposed multitype joint loss function in remote sensing SR.

E. Scene Classification Evaluation of SR-RSSIs

In this section, the performance of scene classification on HR-RSSIs, LR-RSSIs, and SR-RSSIs was discussed. In order to comprehensively compare the differences between various SISR methods in remote sensing scene classification, the experiments were conducted on the SIRI-WHU dataset. First, the LR-RSSIs were generated by downsampling the corresponding HR-RSSIs. Then, SR models were employed to reconstruct the SR-RSSIs from the LR-RSSIs. Finally, the pretrained classification models were adopted to predict the scene category of RSSIs. In this work, three pretrained models, i.e., SqueezeNet, VGG19, and DenseNet161, were utilized for scene classification. Moreover, the ratio of the training set was fixed at 50%, and the remaining images were used for testing.

For a fair comparison, LR-RSSIs, HR-RSSIs, and SR-RSSIs have the same parameter settings for scene classification. The classification results of these images on the three pretrained models are shown in Table V. It can be found that image SR can not only greatly improve the quality of LR-RSSIs, but also promote the application of LR-RSSIs in other remote sensing tasks. Furthermore, compared with other SR-RSSIs, our SR-RSSIs were more suitable for scene classification and achieved the best accuracy. The novel loss function is designed to further improve the perceptual quality of the SR-RSSIs. LGC-GDAN+ has about 1% higher accuracy than LGC-GDAN. It can be concluded that the classification performance of the RSSIs does not depend on the luminance and contrast

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TABLE V
SCENE CLASSIFICATION RESULTS OF LR-RSSIS, HR-RSSIS, AND SR-RSSIS ON THE SIRI-WHU DATASET USING SQUEEZENET, VGG19, AND DENSENET161

SIRI-WHU	HR-RSSI	LR-RSSI	SR-RSSI						
			HAT	VDSR	LGCNet	EEGAN	RCAN	LGC-GDAN	LGC-GDAN +
SqueezeNet	91.83%	82.83%	89.45%	91.42%	91.25%	90.67%	91.75%	91.50%	92.50%
VGG19	94.50%	87.17%	91.27%	93.25%	92.92%	92.08%	93.00%	93.50%	93.83%
DenseNet161	96.17%	92.50%	94.09%	95.33%	94.83%	94.58%	95.67%	95.58%	96.08%

of the images but on the image structure and spatial context. As illustrated in Table V, when using DenseNet161 for the scene classification of LR RSIs, its classification accuracy surpasses that of using SqueezeNet for scene classification of HR RSIs. The main reason is that DenseNet-161 uses dense connections, which are more effective than SqueezeNet in solving the gradient vanishing problem and enhancing feature reuse. In Table, HAT performs poorly for RSI scene classification compared to LGC-GDAN method. This indicates that the natural image SR model may suffer from insufficient adaptability when dealing with RSIs. As can be seen from Table V, the classification accuracy of our SR-RSSIs on VGG19 and DenseNet161 is ∼0.5% lower than that of HR-RSSIs, but the results of our SR-RSSIs on SqueezeNet is even higher than that of HR-RSSIs.

It means that appropriate simplification of the local details of the image is beneficial to RSI scene classification. In general, remote sensing SR should be regarded as an indispensable process for LR-RSSIs to scene classification tasks.

V. CONCLUSION

In this article, the LGC-GDAN is proposed for RSI SR. To solve the complex spatial distribution challenge of RSIs caused by over-head imaging angle, GAS and LAS mechanisms are designed to learn the low and high frequency similarity information of RSIs from multiple perspectives. The edge-assist branch based on the encoder-decoder network is proposed to recover more local details. Furthermore, the generator network is trained on two task-driven loss functions to produce images with different visual appearances. To cope with the problems of small ground objects and complex scenes caused by long-range imaging distance, dual region-level discriminator networks are proposed for adversarial learning. The output of the discriminator is no longer the real/fake label of the entire image but the real/fake label of each local region. The comprehensive experimental results shows that the SR quality of LGC-GDAN outperforms the state-of-the-art SR methods. In addition, our SR method can also improve the performance of LR images in remote sensing scene classification.

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